

TRAINING & TRANSFORMATION PIPELINE

Trainees begin with a two-to-three-month residency at CIP laboratories within the BecA-ILRI Hub in Nairobi for advanced techniques, after initial training organised in Rwanda. The full transformation cycle covers gene construct assembly, Agrobacterium culture maintenance, plasmid validation, explant infection, plant regeneration, molecular characterisation, and greenhouse phenotyping

TARGET VARIETIES & 3R-GENE CONSTRUCT

Farmer-preferred Rwandan varieties Kinigi and Cyerekezo will be transformed with the three-gene construct pCIP99 (GB MN164628.2), containing Rpi-vnt1.1 from Solanum venturii and Rpi-RB and Rpi-blb2 from Solanum bulbocastanum. This 3R-gene stacking delivers broad-spectrum, durable resistance to diverse late blight pathogen strains under field conditions.

CISGENIC APPROACH & REGULATORY PATHWAY

A second construct from TSL stacks Rpi-vnt1, Rpi-amr1, and Rpi-amr3 — all sourced exclusively from the Solanum genus — creating potential for cisgenic classification rather than transgenic, which may simplify the regulatory pathway. The potato ALS gene serves as a selectable marker conferring chlorsulfuron herbicide resistance for streamlined selection.

CONFINED FIELD TRIALS & NATIONAL OWNERSHIP

Resulting transformation events will advance to confined field trials (CFTs) across three existing RABP-CMP sites, with all activities led by trained Rwandan scientists. RAB will carefully select recipient varieties distinct from any pending biotech releases to avoid regulatory complications, as tuber morphology remains the primary field-level identification method.

BioCap
BUILDING RWANDAN CAPACITY for Accelerated Development, Deregulation, and Deployment of Enhanced Crops

THE SCIENCE OF RWANDAN GROWTH

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CIP
INTERNATIONAL POTATO CENTER



BIOTECH POTATO

REDUCING LATE BLIGHT, INCREASING FARMER PROSPERITY

ALLIANCE FOR SCIENCE RWANDA | MICHIGAN STATE UNIVERSITY | DONALD DANFORTH PLANT SCIENCE CENTER | IITA CGIAR

AFRICAN CROPS BY AFRICAN SCIENTISTS, FOR AFRICAN FARMERS

1. THE CHALLENGE

Potato is among Rwanda's highest-value food and cash crops, particularly in the country's highland production zones. However, Late Blight remains the most destructive disease affecting potato production. Farmers often rely on repeated fungicide spraying throughout the growing season to protect their crops. This increases production costs, exposes farmers and the environment to chemical pesticides, and reduces profitability for smallholder households.

Heavy dependence on imported fungicides also contributes to foreign exchange expenditure while increasing production risks during periods of supply disruption.

2. THE BIOTECHNOLOGY SOLUTION

Through the Rwandan Agricultural Biotech Capacity Building programme, Rwanda will deepen its engagement with crop biotechnology by building local laboratory and field capacity for the next generation of biotech crops. For potato, this strengthens national capacity to develop, evaluate, and steward late blight-resistant farmer-preferred potato varieties using modern biotechnology.

The investment document highlights Rwanda’s opportunity to become an African leader in science-based crop improvement and regulation. Potato biotechnology supports that goal by preserving preferred market and food qualities while generating the safety, performance, and stewardship data needed for responsible farmer deployment.

Expected benefits include:

- Reduced fungicide applications
- Lower production costs
- Improved farmer health
- Environmental protection
- Increased yields
- Higher profitability

3. PARTNERSHIP FOR INNOVATION

The Potato Biotechnology Programme is implemented through a strong international partnership:



National leadership for potato research, contained and field evaluation, regulatory coordination, seed system integration, deployment planning, and stewardship after release.

Grantee and overall coordination partner for the Rwanda biotechnology capacity-building investment, as well as a lead technical partner for potato improvement, late blight-resistance strategy, regulatory-quality data support, stewardship, and project governance.

Graduate education, technical mentorship, and institutional capacity building for RAB scientists in molecular breeding, pathology, biosafety, and regulatory science.

Communication and outreach partner for public engagement, farmer field learning, media products, policymaker dialogue, and clear messaging on reduced fungicide use and farmer benefits.

4. BUILDING RWANDA'S FUTURE

Centre of Excellence in Crop Biotechnology: Potato Platform

The investment will establish and strengthen a Centre of Excellence for Crop Biotechnology at RAB, capable of genetic transformation, genome editing, advanced molecular diagnostics, contained evaluation, and regulatory-quality data generation. The potato platform will apply these capabilities to late blight resistance, seed system readiness, stewardship, and evidence-based regulatory review.



